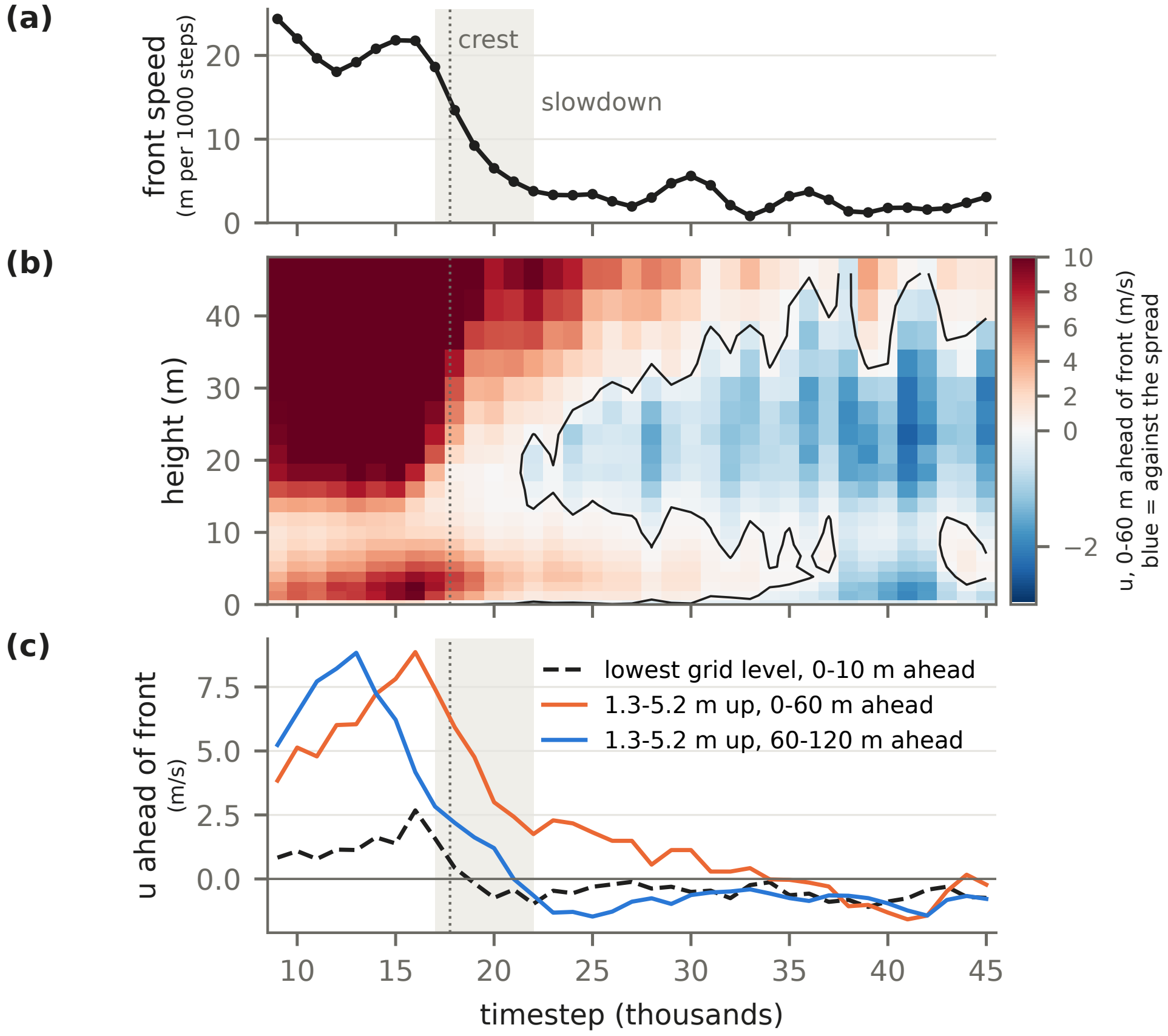
(a)
front speed
(m per 1000 steps)
20
10
0
crest
slowdown
(b)
height (m)
40
30
20
10
0
10
8
6
4
2
0
−2
u, 0-60 m ahead of front (m/s)
blue = against the spread
(c)
u ahead of front
(m/s)
7.5
5.0
2.5
0.0
lowest grid level, 0-10 m ahead
1.3-5.2 m up, 0-60 m ahead
1.3-5.2 m up, 60-120 m ahead
10
15
20
25
30
35
40
45
timestep (thousands)